AUTO AGENT

The Intelligent Orchestration Layer for the Modern Workspace

INTRODUCTION OF THE PROJECT

AutoAgent is an advanced autonomous orchestration system designed to bridge the gap between Large Language Models (LLMs) and real-world execution. While traditional chatbots generate text, AutoAgent generates action.

Powered by Reasoning LLM and the Model Context Protocol the system acts as a universal nervous system for your digital tools. It allows users to control over 800+ applications (including GitHub, HubSpot, and Gmail) through natural language, transforming fragmented workflows into a unified, intelligent stream of execution

OBJECTIVES OF THE PROJECT

Unified Command Center:
The objective is to consolidate interactions with more than 800 SaaS applications into a single natural language interface.

Autonomous Workflow Orchestration:
The system aims to move beyond basic question-and-answer functionality by enabling the agent to plan, execute, and verify multi-step tasks automatically.

Precision Tool Routing:
Another objective is to accurately understand user intent and route it to the correct tool and action using semantic analysis, thereby avoiding incorrect or hallucinated actions.

Real-Time Latency Optimization:
The project aims to achieve fast reasoning and response times using Large Language Models

Secure Integration and Protocol Support:
The objective is to securely integrate with over 800 external applications using the Composio SDK and support context-aware AI-tool interoperability through the Model Context Protocol

PROBLEM SOLVING STATEMENT

Fragmentation and Tab Fatigue

In modern work environments, professionals use multiple tools like IDEs, CRMs, emails, and communication apps. Constantly switching between these applications causes tab fatigue, reduces focus, and decreases productivity.

Execution Gap in AI Tools

Most AI tools can only give suggestions but cannot perform real actions like executing API calls or completing workflows. Users still need to manually carry out tasks, creating a gap between decision and execution.

Proposed Solution

AutoAgent solves this problem by providing a single interface where users give instructions in natural language. It understands user intent and automatically executes tasks across different platforms, reducing context switching and improving efficiency.

TOOLS TO BE USED

Frontend:
HTML5, CSS3, and JavaScript are used to design a simple and responsive user interface. Server-Sent Events are used for real-time data communication.

Backend Framework:
FastAPI with Python is used to develop backend APIs and handle client requests.

AI Engine:
Google Gemini is used for fast processing and understanding of user instructions.

Integration Layer:
Composio SDK and Model Context Protocol are used to connect and manage multiple external tools.

Database:
SQLite is used to store session data and configuration details.

MODULES DESCRIPTION

Chat Module**:**
This module provides the main interface for user interaction. It manages user messages, displays responses, and supports real-time message updates.

Agent Module**:**
This module acts as the core part of the system. It understands user requests, processes instructions, and controls the flow of conversation.

Tool Router Module**:**
This module is used to connect and manage multiple external tools. It selects and executes the appropriate tool based on user requirements.

Sub-Modules**:**
Login and Session Management**:** Manages user sessions and keeps track of individual conversations.
Tool Search**:** Identifies the required tool based on the user's request.
Code Execution**:** Allows the system to run and test code safely when required.

FUTURE SCOPE OF THE PROJECT

The system can be enhanced to support inputs such as images, audio, and video.
A mobile application can be developed for Android and iOS platforms for easy access.
Multiple intelligent agents can be introduced to work together on complex tasks.
The project can be extended to support large organizations with advanced security features.